A SHARP QUARTIC POINTWISE MODICA ESTIMATE FOR THE INFINITY LAPLACIAN

Abstract. Let u be a bounded entire continuous viscosity solution of $\Delta_\infty u = W'(u)$, where the infinity Laplacian is unnormalized and $W \in C^1$ on a neighborhood of the closed range I of u. We prove the sharp estimate $\operatorname{lip} u(x)^4/4 \le W(u(x)) - \min_I W$ at every point, without assuming differentiability of the solution. The proof compares u with increasing radial profiles whose equations contain a strictly signed perturbation. Sliding their initial values produces a smooth touching test away from both the radial vertex and the truncation set. For classical solutions, a second proof uses conservation of the quartic energy along unit-speed gradient curves. We deduce a global Lipschitz bound, an intrinsic distance inequality, and a criterion for constancy when a potential well is attained. Using the cited homogeneous interior $C^{1,\alpha}$ theorem, we give a complete lifting and affine approximation argument that yields $C^{1,\beta}$ regularity and hence the quartic gradient inequality everywhere. One-dimensional profiles prove sharpness. A smooth periodic example attains a zero of an analytic nonnegative potential without being constant. We explain this behavior through the order of vanishing at the well. Smooth radial examples show that equality does not imply one-dimensionality, and compact transitions give nonconstant solutions equal to a well outside a ball.

Draft status. This candidate proof was obtained by our team on October 09, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

We study bounded entire solutions of

$$\Delta_\infty u = W'(u) \qquad \text{in } \mathbb{R}^n, \qquad \Delta_\infty u = \left\langle D^2 u\, Du, Du \right\rangle. \tag{1.1}$$

The operator in (1.1) is the *unnormalized* infinity Laplacian. With the alternative sign convention $-\Delta_\infty u = f$, the potential equation considered here corresponds to $f = -W'(u)$. This sign and the normalization of the operator are essential for the gradient estimate.

1.1. **Background and current research status.** Modica's estimate for bounded entire solutions of the nonlinear Poisson equation $\Delta u = W'(u)$ bounds $|Du|^2/2$ by a nonnegative potential $W(u)$ [9]. Caffarelli, Garofalo and Segàla extended this principle to suitable quasilinear divergence equations [3]. For example, their framework gives the coefficient $(p-1)/p$ in front of $|Du|^p$ for $\Delta_p u = W'(u)$ under the relevant structural hypotheses. Passing formally to $p = \infty$ does not produce the estimate for (1.1): the normalization of the differential equation and of its reaction term changes in that limit.

The absence of divergence form is not itself an obstruction to a gradient–potential inequality. Andrews and Xiong obtain sharp comparisons with one-dimensional profiles for viscosity solutions of isotropic equations [1]. Their stated hypotheses require a strictly positive coefficient in directions orthogonal to a nonzero gradient. The pure infinity Laplacian has zero coefficient in those directions, so that the result used in their introduction does not directly apply to (1.1). Their work nevertheless places profile comparison in the appropriate broader context.

For the infinity Laplacian itself, a close predecessor is the quartic P-function

$$P = \frac{1}{4}|Du|^4 + u \tag{1.2}$$

used by Crasta and Fragalà for $-\Delta_\infty u = 1$ with zero Dirichlet data [5]. Their P-function bounds use convexity, an interior sphere condition, and the additional boundary regularity specified in their Theorem 16. That problem differs from the entire potential equation here. Nonlinear cone comparisons for the normalized infinity Laplacian also appear in the work of Araújo, Mari and Pessoa [2]; the operator and the sign hypotheses in their Lipschitz estimates are different from those of the present theorem.

Homogeneous regularity theory plays a separate role. Evans and Savin established planar $C^{1,\alpha}$ regularity, and Evans and Smart proved everywhere differentiability in arbitrary dimension [6, 7]. For the regularity part of this paper we take the homogeneous interior estimate in the OpenAI preprint dated 4 October 2026 as a theorem [10]. Its precise quantitative form used below is recorded in theorem 1.1.

This review reflects the primary sources examined through 17 October 2026. We have not located in those sources the exact entire-viscosity statement of theorem 1.2. We make no claim of bibliographic priority. The purpose of the paper is to give a self-contained proof of that statement, to separate it from the regularity input, and to establish its sharpness and limitations with explicit examples.

1.2. **The cited homogeneous theorem.** For a vector-valued map F on a set E, write

$$[F]_{\alpha;E} = \sup_{\substack{x,y\in E\\ x\neq y}} \frac{|F(x)-F(y)|}{|x-y|^\alpha}.$$

The following theorem is our external regularity input.

Theorem 1.1 (Homogeneous interior estimate, OpenAI [10, Theorem 1.1])**.** *For every integer $d \geq 3$ there are constants $a_d \in (0, 1/3]$ and $C_d < \infty$, depending only on d, such that every bounded viscosity solution h of $\Delta_\infty h = 0$ in B_1^d belongs to $C^{1,a_d}(B_{1/2}^d)$ and satisfies*

$$\|Dh\|_{L^\infty(B_{1/2}^d)} + [Dh]_{a_d; B_{1/2}^d} \leq C_d \operatorname{osc}_{B_1^d} h. \tag{1.3}$$

Here the exponent may be decreased to lie in $(0, 1/3]$.

We use theorem 1.1 only in section 5. Neither the sharp metric estimate nor its intrinsic distance consequences rely on it. The inhomogeneous regularity conclusion will be proved by a separate argument; it is not asserted as an immediate consequence of the theorem.

1.3. **Main results.** For a continuous function u define its pointwise local slope by

$$\operatorname{lip} u(x) = \limsup_{\substack{y \to x \\ y \neq x}} \frac{|u(y) - u(x)|}{|y - x|}. \tag{1.4}$$

This definition does not require differentiability. If u is differentiable at x, then $\operatorname{lip} u(x) = |Du(x)|$.

Theorem 1.2 (Sharp quartic estimate)**.** *Let $n \geq 1$, let $u \in C(\mathbb{R}^n)$ be bounded, and suppose that (1.1) holds in the viscosity sense. Let $W \in C^1(J)$ for an open interval J containing*

$$I = [m, M], \qquad m = \inf_{\mathbb{R}^n} u, \quad M = \sup_{\mathbb{R}^n} u, \qquad \mu = \min_I W, \quad V = W - \mu. \tag{1.5}$$

Then, at every $x \in \mathbb{R}^n$,

$$\boxed{\frac{1}{4}\big(\operatorname{lip} u(x)\big)^4 \leq V(u(x)) = W(u(x)) - \mu.} \tag{1.6}$$

In particular,

$$\operatorname{Lip}(u; \mathbb{R}^n) \leq L, \qquad L = \Big(4\big(\max_I W - \min_I W\big)\Big)^{1/4}, \tag{1.7}$$

and $|Du|^4/4 \leq W(u) - \mu$ wherever u is differentiable. If $W \geq 0$ on I, the customary form $|Du|^4/4 \leq W(u)$ follows. The coefficient $1/4$ is optimal.

No sign condition on W' is imposed. Only values of W near the closed range are needed. The subtraction of μ makes the conclusion invariant under adding a constant to the potential.

Theorem 1.3 (Everywhere gradient and quantitative regularity)**.** *In the setting of theorem 1.2, take theorem 1.1 as a theorem and set*

$$N = \max\{n + 1, 3\}, \qquad a = a_N, \qquad \beta = \frac{a}{2 + a}, \qquad F = \max_I |W'|. \tag{1.8}$$

Then $u \in C^{1,\beta}(\mathbb{R}^n)$ and

$$\frac{1}{4}|Du(x)|^4 \leq W(u(x)) - \mu \qquad (x \in \mathbb{R}^n). \tag{1.9}$$

If $L, F > 0$, then

$$\|Du\|_{L^\infty(\mathbb{R}^n)} \leq L, \qquad [Du]_{\beta; \mathbb{R}^n} \leq C\, L^{1-3\beta} F^\beta, \tag{1.10}$$

where C depends only on N and the constants in theorem 1.1. If $LF = 0$, then u is constant. The exponent β and the Hölder constant are not claimed to be sharp.

We also prove the intrinsic estimate

$$\int_{\min\{u(x), u(y)\}}^{\max\{u(x), u(y)\}} \frac{ds}{[4V(s)]^{1/4}} \leq |x - y|, \tag{1.11}$$

where the integral may initially be infinite. This yields a criterion for constancy at an attained well and a sharp minimum distance between attained phases. Attaining a well

alone does not imply constancy: for $W(s) = (1 - s^2)^2/4$, the smooth nonconstant solution $u(x) = \cos x_1$ satisfies $W(u(0)) = 0$; see proposition 4.5. For the same potential, the smooth bounded function $u(x) = -\cos|x|$ attains equality in (1.9) everywhere and is not one-dimensional when $n \geq 2$; see proposition 6.5.

1.4. **Outline of the proof.** Section 2 fixes the viscosity signs and records the elementary facts about local slopes and radial functions. Section 3 constructs profiles with positive speed and equation $(q')^2 q'' = W'(q) - \varepsilon$. The profiles are truncated above $\sup u$ and slid by their initial height. A first contact above the desired initial height would be a smooth upper test at an interior point, contradicting the strict perturbation. Letting the positive energy parameter tend to zero and reflecting u gives the sharp slope estimate. Subsection 3.2 gives an independent proof for classical solutions: the quartic energy is constant along a unit-speed gradient curve, and a positive energy would force u to grow without bound along a globally defined curve.

Section 4 proves the intrinsic distance inequality and its consequences at potential wells, including the smooth counterexample showing why attaining a potential well alone does not imply constancy. Section 5 adds one auxiliary variable, uses homogeneous comparison to approximate the lifted solution with an error of order r^2, and combines theorem 1.1 with affine reconstruction. The balance of approximation errors gives $\beta = a/(2 + a)$. Section 6 supplies profiles, sharp examples, and failures of rigidity. Section 7 explains the scope of the estimate for Dirichlet problems and general spatial forcing.

2. Viscosity conventions and elementary facts

2.1. **Signs and test functions.** We write (1.1) in degenerate elliptic form as

$$\mathcal{F}(u, Du, D^2u) = -\left\langle D^2u\, Du, Du \right\rangle + W'(u) = 0. \tag{2.1}$$

We use the standard viscosity convention for this sign [4]. An upper touching test ϕ satisfies $\phi(x_0) = u(x_0)$ and $u - \phi \leq 0$ near x_0; a lower touching test has the reversed local inequality. Thus, if $\phi \in C^2$ touches u from above at x_0, then

$$\Delta_\infty \phi(x_0) \geq W'(u(x_0)); \tag{2.2}$$

if it touches from below, the inequality is reversed. A touching function need only be C^2 in a neighborhood of the touching point. No separate normalized-operator convention is imposed at $D\phi = 0$; the unnormalized operator there is simply zero.

The same convention applies to $\Delta_\infty u = g(x)$: upper tests satisfy $\Delta_\infty \phi \geq g(x_0)$ and lower tests satisfy $\Delta_\infty \phi \leq g(x_0)$. Replacing u by $-u$ changes the potential to $\widetilde{W}(s) = W(-s)$, defined on $-J = \{-s : s \in J\}$, and gives

$$\Delta_\infty(-u) = \widetilde{W}'(-u). \tag{2.3}$$

Indeed, $\widetilde{W}'(s) = -W'(-s)$. If ϕ is an upper test for $-u$ at x_0, then $-\phi$ is a lower test for u, so

$$-\Delta_\infty \phi(x_0) = \Delta_\infty(-\phi)(x_0) \leq W'(u(x_0)).$$

Thus $\Delta_\infty \phi(x_0) \geq \widetilde{W}'(-u(x_0))$. A lower test for $-u$ gives the reversed inequality by the same calculation, proving (2.3) in the viscosity sense.

Lemma 2.1 (From pointwise slopes to a Lipschitz bound)**.** *Let $v \in C(\mathbb{R}^n)$ and suppose that $\operatorname{lip} v(x) \leq K$ with $0 \leq K < \infty$ at every point. Then $|v(x) - v(y)| \leq K|x - y|$ for all x, y.*

Proof. The case $x = y$ is immediate. Otherwise put $d = |x - y| > 0$ and set $h(t) = v(x + t(y - x))$ for $0 \leq t \leq 1$. For every $0 \leq t < 1$, the points at parameters t and $t + s$

have distance sd for $0 < s \le 1 - t$. The local slope bound therefore gives

$$\limsup_{s\downarrow 0} \frac{h(t+s) - h(t)}{s} \le Kd.$$

If $h(1) - h(0) > Kd$, choose k with $Kd < k < h(1) - h(0)$. Then $h(1) - k > h(0)$. The continuous function $h(t) - kt$ therefore has a minimum at some $t_* \in [0, 1)$, since its value at 1 is larger than its value at 0. For $0 < s \le 1 - t_*$ its right increments are nonnegative, so $h(t_* + s) - h(t_*) \ge ks$ and hence

$$\liminf_{s\downarrow 0} \frac{h(t_*+s) - h(t_*)}{s} \ge k > Kd,$$

a contradiction. Interchanging x and y proves the other inequality. □

Lemma 2.2 (Composition of local slopes)**.** *Let v be continuous in a neighborhood of x, and let H be C^1 on an interval containing the values of v near x. At any included endpoint, its derivative is understood one-sided and continuous up to that endpoint. Then*

$$\operatorname{lip}(H \circ v)(x) \le |H'(v(x))| \operatorname{lip} v(x) \tag{2.4}$$

whenever the right-hand local slope is finite.

Proof. For any $\eta > 0$, continuity of H' bounds its absolute value by $|H'(v(x))| + \eta$ on the part of its domain sufficiently close to $v(x)$. Continuity of v puts $v(y)$ in that part for y close to x. The interval between $v(x)$ and $v(y)$ stays there, so the mean value theorem gives

$$|H(v(y)) - H(v(x))| \le (|H'(v(x))| + \eta)|v(y) - v(x)|.$$

Dividing by $|x - y|$ and taking a limsup yields

$$\operatorname{lip}(H \circ v)(x) \le (|H'(v(x))| + \eta) \operatorname{lip} v(x).$$

The assumed finiteness of $\operatorname{lip} v(x)$ permits $\eta \downarrow 0$, including when either $H'(v(x))$ or $\operatorname{lip} v(x)$ is zero. □

2.2. **Radial profiles and scaling.**

Lemma 2.3 (Radial calculation)**.** *Let $q \in C^2((0, R))$, and set $Q(x) = q(|x - x_0|)$. For $0 < r = |x - x_0| < R$, with $e = (x - x_0)/r$, one has*

$$DQ = q'e, \qquad D^2Q = q''e \otimes e + \frac{q'}{r}(\mathrm{Id} - e \otimes e), \qquad \Delta_\infty Q = (q')^2 q''. \tag{2.5}$$

Proof. Here Id denotes the identity matrix and q', q'' are evaluated at r. Away from x_0, $Dr = e$ and $D^2 r = (\mathrm{Id} - e \otimes e)/r$, so the chain rule gives the displayed gradient and Hessian. Since $(\mathrm{Id} - e \otimes e)e = 0$, contraction with $DQ = q'e$ gives $\Delta_\infty Q = (q')^2 q''$. For $n = 1$ the tangential matrix is zero, so the same calculation applies. No assertion at the vertex x_0 is used. □

The scaling used later is also exact in the viscosity sense. Let $u \in C(\Omega)$ solve $\Delta_\infty u = g$ on an open set $\Omega \subset \mathbb{R}^n$, where $g \in C(\Omega)$. For $x_0 \in \mathbb{R}^n$ and $c \in \mathbb{R}$, define

$$v(y) = \frac{u(x_0 + Ry) - c}{A}, \qquad A, R > 0, \tag{2.6}$$

on $\Omega_R = \{y : x_0 + Ry \in \Omega\}$. Then

$$\Delta_\infty v(y) = \frac{R^4}{A^3} g(x_0 + Ry). \tag{2.7}$$

To prove this, let ϕ be an upper test for v at $y_* \in \Omega_R$. Since $A > 0$, the function

$$\psi(x) = c + A\phi((x - x_0)/R)$$

is an upper test for u at $x_* = x_0 + Ry_*$. At these points,

$$D\psi(x_*) = \frac{A}{R} D\phi(y_*), \qquad D^2\psi(x_*) = \frac{A}{R^2} D^2\phi(y_*), \qquad \Delta_\infty \psi(x_*) = \frac{A^3}{R^4} \Delta_\infty \phi(y_*).$$

The upper-test inequality for u therefore gives $\Delta_\infty \phi(y_*) \geq (R^4/A^3) g(x_*)$. A lower test transfers in the same way, with both inequalities reversed. This proves (2.7), also when $D\phi(y_*) = 0$.

3. The sharp quartic estimate

Throughout this section, I, μ, and V have the meaning in (1.5). In particular, $V \geq 0$ on I, but it need not be nonnegative outside that interval.

3.1. **Sliding radial profiles for viscosity solutions.**

Lemma 3.1 (Strict profiles above the solution range)**.** *Fix $x_0 \in \mathbb{R}^n$, $u_0 \in I$, and $c > 0$. There exist $\delta, \varepsilon > 0$ and a positive C^1 function b on $[u_0, M + \delta]$ such that*

$$b(s)^4 = 4(V(s) + c - \varepsilon(s - u_0)), \qquad b(s)^3 b'(s) = W'(s) - \varepsilon. \tag{3.1}$$

Let

$$H(s) = \int_{u_0}^{s} \frac{dt}{b(t)}, \qquad R = H(M + \delta) > 0, \qquad \psi = H^{-1} : [0, R] \to [u_0, M + \delta]. \tag{3.2}$$

For $a \in [u_0, M + \delta]$ define on $\overline{B}_R(x_0)$

$$Q_a(x) = \psi(\min\{H(a) + |x - x_0|, R\}). \tag{3.3}$$

The map $a \mapsto Q_a$ is continuous in the uniform norm on $\overline{B}_R(x_0)$ and is pointwise nondecreasing in a. Moreover,

$$Q_a(x_0) = a, \quad Q_a = M + \delta \text{ on } \partial B_R(x_0), \quad Q_{M+\delta} \equiv M + \delta. \tag{3.4}$$

At every point $x \neq x_0$ with $Q_a(x) < M + \delta$, it is C^2 locally and satisfies

$$\Delta_\infty Q_a(x) = W'(Q_a(x)) - \varepsilon. \tag{3.5}$$

Proof. Choose $\delta > 0$ with $[M, M + \delta] \subset J$ and $V(s) \geq -c/4$ on this interval. Such a choice is possible by continuity and $V(M) \geq 0$. Next choose

$$0 < \varepsilon < \frac{c}{4(M + \delta - u_0)}.$$

For every $s \in [u_0, M + \delta]$, the expression $V(s) + c - \varepsilon(s - u_0)$ is at least $c/2$: on $[u_0, M]$ we have $V \geq 0$, and on $[M, M + \delta]$ we have $V \geq -c/4$, while $\varepsilon(s - u_0) < c/4$ throughout. Thus (3.1) defines a positive C^1 function; its differentiation gives the second identity there. In particular, $b \geq (2c)^{1/4}$, and

$$0 < R \leq \frac{M + \delta - u_0}{(2c)^{1/4}} < \infty.$$

The strict positivity of R follows from $M + \delta - u_0 \geq \delta > 0$.

The expression defining b^4 remains positive on an open interval contained in J and containing $[u_0, M + \delta]$, by continuity and its positive lower bound. The same formulas extend b as a positive C^1 function and H as a C^2 function with $H' = 1/b > 0$. The inverse function theorem therefore gives a C^2 extension of ψ to a neighborhood of $[0, R]$, and

$$\psi' = b(\psi), \qquad \psi'' = b'(\psi) b(\psi). \tag{3.6}$$

The formula (3.3) is jointly continuous in (a, x) on a compact product, which proves continuity in the uniform norm. Monotonicity and (3.4) follow from the monotonicity of H and ψ. At a point specified in the statement,

$$0 < H(a) + |x - x_0| < R,$$

so the truncation is locally inactive and the radius is nonzero. Thus the profile is C^2 there. By lemma 2.3 and (3.6),

$$\Delta_\infty Q_a = b(Q_a)^3 b'(Q_a) = W'(Q_a) - \varepsilon,$$

which is (3.5). □

Lemma 3.2 (Sliding to the prescribed center value)**.** *Suppose u is a bounded continuous viscosity subsolution in the convention (2.2). Fix x_0, set $u_0 = u(x_0)$, and construct the profiles in lemma 3.1. Then*

$$u \le Q_{u_0} \qquad \text{on } \overline{B}_R(x_0). \tag{3.7}$$

Proof. Let

$$\mathcal{A} = \{a \in [u_0, M+\delta] : Q_a \ge u \text{ on } \overline{B}_R(x_0)\}.$$

It is nonempty because $Q_{M+\delta} = M + \delta > M$. The uniform-norm continuity in lemma 3.1 makes $\mathcal{A}$ closed in the compact interval $[u_0, M+\delta]$. It therefore has a smallest element a_*.

Suppose $a_* > u_0$. There must be a point x_* where $Q_{a_*} = u$. Otherwise $Q_{a_*} - u$ would have a strictly positive minimum on the closed ball, say $\gamma > 0$. By uniform-norm continuity, choose $a \in [u_0, a_*)$ sufficiently close to a_* that $\|Q_a - Q_{a_*}\|_\infty < \gamma/2$. Then $Q_a - u \ge \gamma/2$ on the ball, contradicting the minimality of a_*. The contact is not at x_0, because $Q_{a_*}(x_0) = a_* > u_0$. It is not on the sphere, because $Q_{a_*} = M + \delta > u$ there. The contact value is at most M, so it is not on the truncation set.

Consequently Q_{a_*} is a C^2 upper test at an interior point $x_* \ne x_0$, by lemma 3.1; its value belongs to $[u_0, M]$. The viscosity inequality and (3.5) give

$$W'(u(x_*)) \le \Delta_\infty Q_{a_*}(x_*) = W'(Q_{a_*}(x_*)) - \varepsilon = W'(u(x_*)) - \varepsilon.$$

This contradiction proves $a_* = u_0$ and hence (3.7). □

Proof of theorem 1.2, except for sharpness. Fix $x_0 \in \mathbb{R}^n$ and $c > 0$, and set $u_0 = u(x_0)$. Use the profiles of lemma 3.1. Near the center, $Q_{u_0}(x) = \psi(|x - x_0|)$ and $\psi(0) = u_0$. By lemma 3.2, for $0 < r = |x - x_0| < R$,

$$\frac{u(x) - u(x_0)}{r} \le \frac{\psi(r) - \psi(0)}{r}.$$

The right-hand side tends to $\psi'(0)$ as $r \downarrow 0$ by lemma 3.1. Consequently,

$$\limsup_{x \to x_0} \frac{u(x) - u(x_0)}{|x - x_0|} \le \psi'(0) = b(u_0) = \left[4(V(u_0) + c)\right]^{1/4}. \tag{3.8}$$

This limit is taken for each fixed c; the dependence of R on c therefore causes no difficulty. Although δ and ε also depend on c, the initial speed $b(u_0)$ is independent of both. Let $c \downarrow 0$.

Apply the same argument to $-u$ and $\widetilde{W}(s) = W(-s)$ on $[-M, -m]$. Its minimum is again μ, and (2.3) applies. Thus

$$\limsup_{x \to x_0} \frac{u(x_0) - u(x)}{|x - x_0|} \le [4V(u_0)]^{1/4}.$$

The absolute-value quotient is the maximum of this signed quotient and the one in (3.8). The limsup of the maximum of two functions is the maximum of their limsups. Combining the two bounds therefore gives

$$\operatorname{lip} u(x_0) \le [4V(u_0)]^{1/4}.$$

Taking fourth powers gives (1.6), including when $V(u_0) = 0$.

Since $V \le \max_I W - \min_I W$ on I, lemma 2.1 gives (1.7). At a differentiability point, the definition of the derivative gives $\operatorname{lip} u = |Du|$. If $W \ge 0$ on I, then $\mu \ge 0$, proving the customary estimate. Sharpness is proved in proposition 6.3 below. □

Remark 3.3 (Why no monotonicity of the reaction is needed). The preceding argument does not compare two arbitrary solutions of a reaction equation. At the first contact the values of the solution and the test function agree exactly. Their reaction terms are therefore identical, and the perturbation $-\varepsilon$ supplies the contradiction. No monotonicity of W' or comparison theorem for a sign-changing source has been used.

3.2. **A gradient-flow proof for classical solutions.** For C^2 solutions, conservation of the quartic energy gives a direct proof of the pointwise estimate. We retain the normalization $V = W - \mu$, so that the conclusion agrees with (1.6).

Proposition 3.4 (Classical pointwise estimate)**.** *Let $n \geq 1$, and let $u \in C^2(\mathbb{R}^n)$ be a bounded classical solution of (1.1). Suppose that $W \in C^1(J)$ for an open interval J containing $I = [\inf_{\mathbb{R}^n} u, \sup_{\mathbb{R}^n} u]$. With μ and V as in (1.5), one has*

$$\frac{1}{4}|Du(x)|^4 \leq V(u(x)) = W(u(x)) - \mu \qquad (x \in \mathbb{R}^n). \tag{3.9}$$

In particular, if $W \geq 0$ on I, then $|Du|^4/4 \leq W(u)$ everywhere.

Proof. Define

$$P(x) = \frac{1}{4}|Du(x)|^4 - V(u(x)).$$

Since $u \in C^2$ and $V \in C^1(J)$, the function P is C^1 on $\mathbb{R}^n$. Using $D(|Du|^2) = 2D^2u\, Du$ and $V' = W'$, we obtain

$$DP = |Du|^2 D^2u\, Du - W'(u)Du.$$

The equation therefore gives

$$DP \cdot Du = |Du|^2(\Delta_\infty u - W'(u)) = 0. \tag{3.10}$$

Suppose for contradiction that $P(x_0) = p > 0$ at some point. On the open set $\Omega = \{x \in \mathbb{R}^n : P(x) > 0\}$, nonnegativity of V on I gives

$$|Du(x)|^4 = 4(V(u(x)) + P(x)) > 0.$$

Consequently $X = Du/|Du|$ is well defined and C^1 on Ω, hence locally Lipschitz there. The local existence and uniqueness theorem for ordinary differential equations gives a unique maximal forward solution $\gamma : [0, T) \to \Omega$, with $T \in (0, \infty]$, of

$$\dot\gamma(t) = X(\gamma(t)) = \frac{Du(\gamma(t))}{|Du(\gamma(t))|}, \qquad \gamma(0) = x_0.$$

The chain rule and (3.10) imply

$$\frac{d}{dt}P(\gamma(t)) = \frac{DP(\gamma(t)) \cdot Du(\gamma(t))}{|Du(\gamma(t))|} = 0.$$

Thus, for every $0 \leq t < T$,

$$P(\gamma(t)) = p, \qquad |Du(\gamma(t))|^4 = 4(V(u(\gamma(t))) + p) \geq 4p.$$

In particular, $|Du(\gamma(t))| \geq c$, where $c = (4p)^{1/4} > 0$.

To prove that $T = \infty$, suppose $T < \infty$. The unit-speed identity $|\dot\gamma| = 1$ gives

$$|\gamma(t) - \gamma(s)| \leq |t - s| \qquad (0 \leq s, t < T).$$

Hence $\gamma(t)$ is Cauchy as $t \uparrow T$ and has a limit $x_* \in \mathbb{R}^n$. By continuity, $P(x_*) = p > 0$, so $x_* \in \Omega$ and $|Du(x_*)| \geq c$. Local existence at x_* provides a solution $\eta : [0, \delta) \to \Omega$ with $\eta(0) = x_*$ for some $\delta > 0$. Set $\gamma(T) = x_*$ and continue it by $\gamma(t) = \eta(t - T)$ for $T \leq t < T + \delta$. Since $\dot\gamma(t) = X(\gamma(t)) \to X(x_*) = \dot\eta(0)$ as $t \uparrow T$, the continuation is C^1 and solves the same equation across T. This contradicts maximality and proves the claim. In particular, no global bound on Du is needed to continue the unit-speed trajectory.

Finally,

$$\frac{d}{dt}u(\gamma(t)) = Du(\gamma(t)) \cdot \dot{\gamma}(t) = |Du(\gamma(t))| \geq c \qquad (t \geq 0).$$

Integrating gives $u(\gamma(t)) \geq u(x_0)+ct$ for every $t \geq 0$, contradicting $\sup_{\mathbb{R}^n} u < \infty$. Therefore $P \leq 0$ on $\mathbb{R}^n$, which is (3.9). If $W \geq 0$ on I, then $\mu \geq 0$, and the customary estimate follows. □

Remark 3.5 (The role of smoothness). The preceding proof uses $u \in C^2$ to differentiate P and to obtain a locally Lipschitz vector field on $\{P > 0\}$. For nonsmooth viscosity solutions, conservation along a gradient curve does not follow from the displayed calculation. Crasta and Fragalà discuss the associated absolute-continuity issue [5]. The radial-profile proof above uses smooth test functions only and requires no gradient-flow chain rule.

4. Intrinsic distance and behavior at potential wells

Definition 4.1. For $s, t \in I$, define

$$d_V(s,t) = \int_{\min\{s,t\}}^{\max\{s,t\}} \frac{d\tau}{[4V(\tau)]^{1/4}}. \tag{4.1}$$

The integrand is interpreted as $+\infty$ where $V = 0$, and the integral as an extended nonnegative Lebesgue integral. Set $d_V(s, s) = 0$.

Theorem 4.2 (Intrinsic distance bound)**.** *Under the assumptions of theorem 1.2,*

$$d_V(u(x), u(y)) \leq |x - y| \qquad \text{for all } x, y \in \mathbb{R}^n. \tag{4.2}$$

This conclusion does not use theorem 1.1.

Proof. If $m = M$, then u is constant and the assertion follows from $d_V(m, m) = 0$. Assume $m < M$. For $\eta > 0$, let

$$H_\eta(s) = \int_m^s [4(V(\tau) + \eta)]^{-1/4}\, d\tau, \qquad s \in I.$$

The integrand is positive and continuous on I, so H_η is C^1 there, with continuous one-sided derivatives at the endpoints, and $H'_\eta(s) = [4(V(s) + \eta)]^{-1/4}$. Consequently the endpoint version of lemma 2.2, together with theorem 1.2, gives

$$\operatorname{lip}(H_\eta \circ u)(x) \leq \left(\frac{V(u(x))}{V(u(x)) + \eta}\right)^{1/4} \leq 1.$$

Thus $H_\eta \circ u$ is globally 1-Lipschitz by lemma 2.1. Since H_η is increasing, this gives

$$\int_{\min\{u(x),u(y)\}}^{\max\{u(x),u(y)\}} [4(V(s) + \eta)]^{-1/4}\, ds \leq |x - y|.$$

Choose a sequence $\eta_j \downarrow 0$. At every $s \in I$, $[4(V(s) + \eta_j)]^{-1/4}$ increases to $[4V(s)]^{-1/4}$, with limit $+\infty$ when $V(s) = 0$. Monotone convergence on the fixed interval between $u(x)$ and $u(y)$ proves (4.2). In particular, the integral between any two attained values is finite, irrespective of the zero set of V. □

Corollary 4.3 (Constancy at a well under an integral condition)**.** *Under the assumptions of theorem 1.2, suppose that $s_0 \in I$, $V(s_0) = 0$, and $u(x_0) = s_0$. If*

$$\int_{s_0}^{s_0+\eta} V(s)^{-1/4}\, ds = +\infty \tag{4.3}$$

for every sufficiently small $\eta > 0$ with $s_0 + \eta \leq M$, then $u \leq s_0$. If instead

$$\int_{s_0-\eta}^{s_0} V(s)^{-1/4}\, ds = +\infty$$

for every sufficiently small $\eta > 0$ with $s_0 - \eta \geq m$, then $u \geq s_0$. If the corresponding condition holds on each side that lies in I, then $u \equiv s_0$. In particular, the two-sided conclusion holds if

$$V(s) \leq C|s - s_0|^p \quad \text{for } s \in I \text{ near } s_0, \qquad C > 0, \quad p \geq 4. \tag{4.4}$$

Proof. If $u(y) > s_0$, choose $0 < \eta < u(y) - s_0$ small enough for (4.3) to apply. Since $u(y) \leq M$, this choice also satisfies $s_0 + \eta \leq M$, and

$$d_V(u(x_0), u(y)) \geq 4^{-1/4} \int_{s_0}^{s_0+\eta} V(s)^{-1/4}\, ds = +\infty.$$

This contradicts theorem 4.2. Reflection proves the lower-side statement. Under (4.4), the integrand is bounded below by a positive multiple of $|s - s_0|^{-p/4}$, whose integral at s_0 diverges. If a side does not lie in I, there are no values on that side to exclude. □

Corollary 4.4 (Growth from a well of order below four)**.** *Under the assumptions of theorem 1.2, suppose that $u(x_0) = s_0$, $V(s_0) = 0$, and, for some $C, \delta > 0$ and $0 < p < 4$,*

$$V(s) \leq C(s - s_0)^p \qquad (s_0 \leq s \leq s_0 + \delta)$$

on the part of this interval contained in I. If $s_0 \leq u(x) \leq s_0 + \delta$, then

$$u(x) - s_0 \leq \Big[(1 - p/4)(4C)^{1/4}|x - x_0|\Big]^{4/(4-p)}. \tag{4.5}$$

The analogous assertion holds on the lower side of a well.

Proof. When $u(x) > s_0$, the potential bound gives

$$[4V(s)]^{-1/4} \geq (4C)^{-1/4}(s - s_0)^{-p/4} \qquad (s_0 < s \leq u(x)),$$

where the inequality also holds in the extended sense at zeros of V. Integrating and using theorem 4.2 yields

$$|x - x_0| \geq (4C)^{-1/4} \int_{s_0}^{u(x)} (s - s_0)^{-p/4}\, ds = \frac{(4C)^{-1/4}}{1 - p/4}(u(x) - s_0)^{1-p/4}.$$

Since $1 - p/4 > 0$, rearranging proves (4.5). The case $u(x) = s_0$ is immediate, and reflection gives the lower-side version. □

Proposition 4.5 (A smooth nonconstant solution attaining a well)**.** *For every $n \geq 1$, consider the standard double-well potential*

$$W(s) = \frac{1}{4}(1 - s^2)^2 \tag{4.6}$$

and the function

$$u(x) = \cos x_1 \qquad (x \in \mathbb{R}^n). \tag{4.7}$$

Then u is a smooth bounded entire solution of $\Delta_\infty u = W'(u)$, with $\mu = 0$, and satisfies

$$\frac{1}{4}|Du(x)|^4 = W(u(x)) \qquad (x \in \mathbb{R}^n).$$

The solution attains both zeros of W and is nonconstant. In particular, $W(u(x_0)) = 0$ at one point does not imply constancy, even when both u and the nonnegative potential W are analytic.

Proof. Since $u_{x_1} = -\sin x_1$ and $u_{x_1x_1} = -\cos x_1$, while all derivatives in other coordinate directions vanish, we have

$$\begin{aligned} \Delta_\infty u &= -\sin^2 x_1 \cos x_1 = u^3 - u = W'(u), \\ \frac{1}{4}|Du|^4 &= \frac{1}{4}\sin^4 x_1 = W(u). \end{aligned} \tag{4.8}$$

These identities hold at every point. To verify the viscosity inequalities, an upper test ϕ for the smooth function u has $D\phi = Du$ and $D^2\phi \geq D^2u$ at the touching point, so

$\Delta_\infty \phi \geq \Delta_\infty u = W'(u)$. A lower test gives the reversed inequality. This includes points where $Du = 0$, at which both sides of the classical equation are zero. The range is $[-1, 1]$, and $W \geq 0$ with $W(\pm 1) = 0$, so $\mu = 0$. Since $u(0) = 1$ and $u(\pi e_1) = -1$, where e_1 is the first coordinate unit vector, u is nonconstant and attains both wells. Both the cosine and the polynomial potential are analytic. □

Remark 4.6 (Vanishing order at an attained well). If $u(x_0) = s_0$ and $V(s_0) = 0$, theorem 1.2 gives $\operatorname{lip} u(x_0) = 0$, hence

$$u(x) - u(x_0) = o(|x - x_0|) \qquad (x \to x_0).$$

In particular, u is differentiable at x_0 with $Du(x_0) = 0$, without using theorem 1.1. This local conclusion does not imply constancy. If, for some $C > 0$, $V(s) \leq C|s - s_0|^2$ for $s \in I$ near s_0, applying corollary 4.4 with $p = 2$ on each available side gives

$$|u(x) - s_0| \leq \frac{\sqrt{C}}{2}|x - x_0|^2 \qquad \text{for } x \text{ sufficiently close to } x_0.$$

Continuity of u ensures that these nearby values lie in the neighborhood where the potential bound holds. This estimate permits quadratic departure from the well. For (4.7), $1 - u(te_1) = t^2/2 + O(t^4)$, which also realizes the growth exponent in corollary 4.4 for $p = 2$.

The potential in (4.6) has degree four as a polynomial, but its zeros at ± 1 have order two. More precisely,

$$\frac{W(s)}{|s - s_0|^2} = \frac{(s + s_0)^2}{4} \longrightarrow 1 \qquad (s \to s_0, \quad s_0 \in \{-1, 1\}).$$

Thus $W(s)^{-1/4}$ is comparable to $|s - s_0|^{-1/2}$ near each well and is locally integrable. The integral hypothesis of corollary 4.3 therefore fails in this example. Additional smoothness of the potential does not change this integrability.

For comparison, $\widehat{W}(s) = c(1 - s^2)^4$ with $c > 0$ has zeros of order four at ± 1. Near either zero, $\widehat{W}(s) \leq C|s - s_0|^4$. For a bounded entire solution of $\Delta_\infty u = \widehat{W}'(u)$ attaining either well, nonnegativity of $\widehat{W}$ gives $\min_I \widehat{W} = 0$, so its normalized potential is $V = \widehat{W}$. Thus corollary 4.3 applies and the solution is constant. For $c = 1/4$, the tanh solution in proposition 6.3 approaches these wells only at infinity. This distinction concerns the order of vanishing at a well, rather than the degree of the potential as a polynomial.

5. Pointwise gradient regularity from homogeneous estimates

We now use theorem 1.1. The sharp estimate and its intrinsic consequences were proved for continuous viscosity solutions. This order avoids any circular use of differentiability in the sharp estimate.

5.1. **An exact viscosity lifting lemma.**

Lemma 5.1 (Addition of a smooth function in a new variable)**.** *Let Ω be an open subset of a Euclidean space and $J_0 \subset \mathbb{R}$ an open interval. Suppose $v \in C(\Omega)$ solves $\Delta_\infty v = g(y)$ in the viscosity sense, where $g \in C(\Omega)$, and let $w \in C^2(J_0)$. Then*

$$U(y, t) = v(y) + w(t) \quad \textit{solves} \quad \Delta_\infty U = g(y) + (w'(t))^2 w''(t) \quad \textit{in } \Omega \times J_0. \tag{5.1}$$

Proof. Let $\phi \in C^2$ touch U from above at (y_0, t_0). Restricting the contact to the vertical line $y = y_0$ gives $\phi_t(y_0, t_0) = w'(t_0)$. Set $p = D_y\phi(y_0, t_0)$ and $k = w'(t_0)$, so that $D\phi(y_0, t_0) = (p, k)$. If $p \neq 0$, put $b = kp/|p|^2$ and consider

$$\zeta(y) = \phi(y, t_0 + b \cdot (y - y_0)) - w(t_0 + b \cdot (y - y_0)).$$

After shrinking the neighborhood of y_0, this affine graph stays inside the neighborhood of contact for ϕ. Thus $\zeta \geq v$ there, with equality at y_0, and ζ is a C^2 upper test. Its first

derivative is $D\zeta(y_0) = D_y\phi + (\phi_t - w')b = p$, and, with all derivatives of ϕ evaluated at (y_0, t_0),

$$D^2\zeta(y_0) = D^2_{yy}\phi + D_y\phi_t \otimes b + b \otimes D_y\phi_t + (\phi_{tt} - w''(t_0))b \otimes b.$$

Since $b \cdot p = k$, contraction with p gives

$$p^T D^2\zeta(y_0)p = (p,k)^T D^2\phi(y_0,t_0)(p,k) - k^2 w''(t_0). \tag{5.2}$$

Thus the mixed derivatives are retained in the direction (p, k). The upper-test inequality for v, together with $D\zeta(y_0) = p$, now yields $\Delta_\infty\phi \geq g(y_0) + k^2 w''(t_0)$.

If $p = 0$, the function $y \mapsto \phi(y, t_0) - w(t_0)$ is a horizontal upper test for v with zero gradient, giving $0 \geq g(y_0)$. The vertical upper contact gives $\phi_{tt} \geq w''(t_0)$. Hence

$$\Delta_\infty\phi = k^2\phi_{tt} \geq k^2 w''(t_0) \geq g(y_0) + k^2 w''(t_0).$$

These upper-test calculations also cover $k = 0$. For a lower test ϕ for U, the function $-\phi$ is an upper test for $-U = (-v) + (-w)$. Negating tests as in (2.3) shows that $-v$ solves $\Delta_\infty(-v) = -g$. Applying the upper-test argument just proved to $(-v, -w)$ therefore gives

$$-\Delta_\infty\phi(y_0, t_0) \geq -g(y_0) - k^2 w''(t_0),$$

which is the required lower-test inequality. This proves (5.1) in all cases. □

Taking $w = 0$ in this lemma shows that adjoining unused coordinates preserves the equation. Below the horizontal dimension is $N - 1$, with N as in (1.8).

5.2. **Homogeneous replacement of a lifted function.** We use the classical homogeneous Dirichlet theorem and comparison principle [8]: Lipschitz boundary data on a ball have a unique continuous infinity-harmonic extension, and homogeneous viscosity subsolutions and supersolutions with ordered boundary values compare inside the ball. This is the only comparison principle needed in this section.

For $|d| \leq 1$ and $|t| \leq 1$, define

$$w_d(t) = \int_0^t (8 + 3ds)^{1/3}\, ds. \tag{5.3}$$

For each $|d| \leq 1$, the same formula defines a smooth function on $(-8/3, 8/3)$, so the lifting lemma applies on a neighborhood of the closed vertical interval $[-1, 1]$. Differentiation in t and d, with the substitution $s = t\tau$ in the parameter derivative, gives

$$\begin{gathered}(w_d')^2 w_d'' = d, \qquad w_0(t) = 2t, \\ \partial_d w_d(t) = \int_0^t s(8 + 3ds)^{-2/3}\, ds = t^2 \int_0^1 \tau(8 + 3dt\tau)^{-2/3}\, d\tau.\end{gathered} \tag{5.4}$$

The factor t^2 also makes the sign explicit for $t < 0$. Since $8 + 3dt\tau \geq 5$, this implies

$$0 \leq \partial_d w_d(t) \leq \frac{t^2}{2 \cdot 5^{2/3}} \leq 1 \qquad (|d|, |t| \leq 1). \tag{5.5}$$

Lemma 5.2 (Uniform replacement error)**.** *Suppose v is Lipschitz on $\overline{B}_1^{N-1}$, $|v| \leq 1$, and $\Delta_\infty v = g$ in B_1^{N-1} with $g \in C(B_1^{N-1})$ and $|g| \leq r \leq 1$. Let h be the infinity-harmonic function in B_1^N with boundary values $U(y,t) = v(y) + 2t$. Then*

$$\|h - U\|_{L^\infty(B_1^N)} \leq r, \qquad \operatorname{osc}_{B_1^N} h \leq 6. \tag{5.6}$$

Proof. The boundary datum $U(y,t) = v(y) + 2t$ is Lipschitz on ∂B_1^N, so the homogeneous Dirichlet theorem supplies h. Set

$$S(y,t) = v(y) + w_r(t) - r, \qquad T(y,t) = v(y) + w_{-r}(t) + r.$$

By lemma 5.1, $\Delta_\infty S = g + r \geq 0$ and $\Delta_\infty T = g - r \leq 0$ in the viscosity sense. Thus S and T are, respectively, a homogeneous subsolution and supersolution with the sign convention of (2.1). By (5.5),

$$U - r \leq S \leq U \leq T \leq U + r.$$

Both S and T are continuous on $\overline{B}_1^N$, and their boundary values bracket those of h. Homogeneous comparison gives $S \leq h \leq T$, proving the first assertion. The boundary values of h lie between -3 and 3; comparison with constants proves the oscillation bound. □

5.3. **Affine approximation at every center.**

Lemma 5.3 (Reconstruction from uniform affine errors)**.** *Suppose $v \in C(\mathbb{R}^k)$, $0 < \beta < 1$, and that there are fixed $K, P, \rho_0 > 0$ such that for every $x \in \mathbb{R}^k$ and $0 < \rho \leq \rho_0$ one can choose $p_{x,\rho}$ with*

$$|p_{x,\rho}| \leq P, \qquad \sup_{|z|\leq\rho} |v(x+z) - v(x) - p_{x,\rho}\cdot z| \leq K\rho^{1+\beta}.$$

Then v is continuously differentiable on $\mathbb{R}^k$, $\|Dv\|_\infty \leq P$, and

$$[Dv]_{\beta;\mathbb{R}^k} \leq C_\beta K + C_\beta P \rho_0^{-\beta}. \tag{5.7}$$

Proof. Fix x and let $\rho_j = 2^{-j}\rho_0$. Compare consecutive affine approximations at $z = (\rho_j/2)e$, where e is any unit vector. This gives

$$|(p_{x,\rho_j} - p_{x,\rho_{j+1}})\cdot e| \leq 2K(1 + 2^{-1-\beta})\rho_j^\beta \leq 4K\rho_j^\beta.$$

Taking the supremum over e and summing the resulting geometric series shows that p_{x,ρ_j} converges to a vector p_x and

$$|p_x - p_{x,\rho_j}| \leq \sum_{i=j}^{\infty} 4K\rho_i^\beta = \frac{4K}{1-2^{-\beta}}\rho_j^\beta.$$

If $0 < |z| \leq \rho_0$, choose j with $\rho_{j+1} < |z| \leq \rho_j$. The affine approximation at radius ρ_j, together with $\rho_j < 2|z|$, then implies

$$\begin{aligned} |v(x+z) - v(x) - p_x\cdot z| &\leq K\rho_j^{1+\beta} + \frac{4K}{1-2^{-\beta}}\rho_j^\beta |z| \\ &\leq C_\beta K|z|^{1+\beta}. \end{aligned} \tag{5.8}$$

Hence v is differentiable at every x and $Dv(x) = p_x$. The bound $|Dv(x)| \leq P$ follows by passing to the limit.

Write $E_x(z) = v(x+z) - v(x) - Dv(x)\cdot z$. For $x \neq y$, $d = |x-y| \leq \rho_0/2$, and any unit vector e, subtraction of the expansions at x and y gives the exact identity

$$d\,(Dv(y) - Dv(x))\cdot e = E_x(y - x + de) - E_x(y-x) - E_y(de).$$

All three increments have length at most $2d \leq \rho_0$; a zero increment contributes $E_x(0) = 0$. Using (5.8), dividing by d, and taking the supremum over e gives $|Dv(y) - Dv(x)| \leq C_\beta K d^\beta$. When $d > \rho_0/2$, the bound $|Dv(y) - Dv(x)| \leq 2P$ gives

$$\frac{|Dv(y) - Dv(x)|}{d^\beta} \leq 2^{1+\beta} P \rho_0^{-\beta}.$$

The constants are independent of x and y, so these estimates prove continuity of Dv and (5.7) globally. □

Proposition 5.4 (Normalized inhomogeneous regularity)**.** *Take theorem 1.1 as a theorem. Let $v \in C(\mathbb{R}^{N-1})$ be globally 1-Lipschitz and solve $\Delta_\infty v = g$, where $g \in C(\mathbb{R}^{N-1})$ and $|g| \leq 1$. Then v is continuously differentiable, its gradient is globally β-Hölder with $\beta = a/(2+a)$, and*

$$[Dv]_{\beta;\mathbb{R}^{N-1}} \leq C. \tag{5.9}$$

The constant depends only on the homogeneous estimate in dimension N.

Proof. Fix a center x and $0 < r \le 1$. On B_1^{N-1} set

$$v_{x,r}(y) = \frac{v(x+ry) - v(x)}{r}.$$

Its absolute value is at most 1, and $\Delta_\infty v_{x,r} = rg(x+ry)$ by (2.7). Let $h_{x,r}$ be the replacement from lemma 5.2. By theorem 1.1,

$$\|Dh_{x,r}\|_{L^\infty(B^N_{1/2})} + [Dh_{x,r}]_{a;B^N_{1/2}} \le C. \tag{5.10}$$

Write $\xi_{x,r} = D_y h_{x,r}(0,0)$, so $|\xi_{x,r}| \le C$. For $|z| \le \rho < r/2$, the entire segment $\{(\tau z/r, 0) : 0 \le \tau \le 1\}$ lies in $B^N_{1/2}$. The fundamental theorem of calculus along this segment gives

$$h_{x,r}(z/r,0) - h_{x,r}(0,0) - \xi_{x,r}\cdot(z/r) = \int_0^1 \left[D_y h_{x,r}(\tau z/r,0) - D_y h_{x,r}(0,0)\right]\cdot(z/r)\,d\tau.$$

Consequently (5.10) implies

$$|h_{x,r}(z/r,0) - h_{x,r}(0,0) - \xi_{x,r}\cdot(z/r)| \le \frac{C}{1+a}(|z|/r)^{1+a}.$$

The replacement error is at most r at both $(z/r,0)$ and $(0,0)$. More precisely,

$$\begin{aligned} v(x+z) - v(x) - \xi_{x,r}\cdot z = {} & r\left[v_{x,r}(z/r) - h_{x,r}(z/r,0)\right] \\ & - r\left[v_{x,r}(0) - h_{x,r}(0,0)\right] \\ & + r\left[h_{x,r}(z/r,0) - h_{x,r}(0,0) - \xi_{x,r}\cdot(z/r)\right]. \end{aligned}$$

The triangle inequality therefore yields

$$\sup_{|z|\le\rho} |v(x+z) - v(x) - \xi_{x,r}\cdot z| \le 2r^2 + C\rho^{1+a}r^{-a}. \tag{5.11}$$

In particular, no equality between $h_{x,r}(0,0)$ and $v_{x,r}(0)$ has been assumed.

Set

$$\theta = \frac{1+a}{2+a}, \qquad r = \rho^\theta. \tag{5.12}$$

For $0 < \rho \le \rho_0 := 1/16$, one has $r \le 1$ and $\rho/r = \rho^{1/(2+a)} \le 16^{-3/7} < 1/2$, since $0 < a \le 1/3$. Thus the Taylor estimate applies at every such radius. Moreover,

$$r^2 = \rho^{1+\beta}, \qquad \rho^{1+a}r^{-a} = \rho^{1+\beta}.$$

For every center x and every such ρ, define $p_{x,\rho} = \xi_{x,\rho^\theta}$. These vectors satisfy $|p_{x,\rho}| \le C$ and

$$\sup_{|z|\le\rho} |v(x+z) - v(x) - p_{x,\rho}\cdot z| \le C\rho^{1+\beta}. \tag{5.13}$$

The slope and error constants are independent of both x and ρ. Thus lemma 5.3 applies with fixed K, P and $\rho_0 = 1/16$, and proves (5.9). □

5.4. **The quantitative conclusion at the natural scale.**

Proof of theorem 1.3. First assume $L, F > 0$, and set

$$\ell = \frac{L^3}{F}, \qquad v(y) = \frac{u(\ell y) - u(0)}{L\ell}. \tag{5.14}$$

By theorem 1.2, v is globally 1-Lipschitz. By (2.7), its source term is

$$\Delta_\infty v(y) = \frac{\ell}{L^3} W'(u(\ell y)),$$

which has absolute value at most 1. For $n \ge 2$, apply proposition 5.4 directly. If $n = 1$, apply it to $\widehat{v}(y_1, y_2) = v(y_1)$, whose equation is preserved by lemma 5.1, and restrict its $C^{1,\beta}$ conclusion and gradient estimate to the line $y_2 = 0$. In either case, v is $C^{1,\beta}$ and $Du(\ell y) = L\,Dv(y)$. Consequently,

$$[Du]_{\beta;\mathbb{R}^n} = L\ell^{-\beta}[Dv]_{\beta;\mathbb{R}^n} \le CL\ell^{-\beta} = CL^{1-3\beta}F^\beta.$$

Everywhere differentiability now turns (1.6) into (1.9), and the sharp supremum bound follows.

If $L = 0$, (1.7) already implies that u is constant. If $F = 0$, then $W' = 0$ on I, so W is constant there and again $L = 0$. This covers all degenerate cases. □

Remark 5.5 (The dimension and the role of the slice). The exponent used above is the homogeneous exponent in dimension N, not automatically the exponent in the original dimension n. Also, the slice $h_{x,r}(\cdot, 0)$ is used only to obtain an affine approximation. It is not claimed to solve a homogeneous equation in the horizontal variables. The lifting proof explicitly retains the mixed second derivatives in (5.2).

6. Sharpness, calibrated profiles, and failures of rigidity

6.1. One-dimensional calibrated profiles.

Proposition 6.1 (Profiles attaining the estimate). *Let $A < B$, let $W \in C^1$ on a neighborhood of $[A, B]$, and suppose $W \geq 0$ on that neighborhood,*

$$W(A) = W(B) = 0, \qquad W > 0 \text{ on } (A, B).$$

Fix $s_* \in (A, B)$ and define

$$\Phi(s) = \int_{s_*}^{s} [4W(t)]^{-1/4}\, dt, \qquad A < s < B. \tag{6.1}$$

Let T_- and T_+ be the extended endpoint limits of Φ. On (T_-, T_+) define $q = \Phi^{-1}$; if an endpoint is finite, set $q(t) = A$ for $t \leq T_-$ or $q(t) = B$ for $t \geq T_+$, respectively, including the joining point. Then $q \in C^1(\mathbb{R})$, and for every unit vector $e \in \mathbb{R}^n$,

$$u(x) = q(e \cdot x) \tag{6.2}$$

is a bounded entire viscosity solution of (1.1) satisfying

$$\frac{1}{4}|Du|^4 = W(u) \quad \text{everywhere.} \tag{6.3}$$

If T_- and T_+ are finite, its transition length is

$$T_+ - T_- = \int_A^B [4W(s)]^{-1/4}\, ds. \tag{6.4}$$

Proof. Since $W > 0$ on (A, B), the function Φ is C^2 with $\Phi' > 0$ and maps (A, B) bijectively onto (T_-, T_+). The normalization $\Phi(s_*) = 0$ gives $T_- < 0 < T_+$. Its inverse tends to A as $t \downarrow T_-$ and to B as $t \uparrow T_+$, with the same assertions at infinite endpoints. On the open transition interval, the inverse function theorem gives $q \in C^2$ and

$$q' = [4W(q)]^{1/4} > 0.$$

Differentiating $(q')^4 = 4W(q)$ and dividing by the positive q' gives $(q')^2 q'' = W'(q)$.

At each finite endpoint, q tends to the corresponding well and q' to zero. The constant extension is therefore C^1 at that endpoint: the one-sided difference quotient of q is a value of q' between that endpoint and the nearby transition point, by the mean value theorem, and hence tends to zero. The derivative on the constant side is also zero. Nonnegativity of W on a neighborhood makes both endpoints local minima, so $W'(A) = W'(B) = 0$.

For (6.2), on the transition region we have

$$Du = q'(e \cdot x)e, \qquad D^2u = q''(e \cdot x)e \otimes e, \qquad \Delta_\infty u = (q')^2 q'' = W'(u).$$

The equation also holds classically on every open constant region. The classical-to-viscosity test calculation in the proof of proposition 4.5 applies on these C^2 regions. At a finite joining hyperplane, u is differentiable with $Du = 0$. Every touching test therefore has gradient zero, so its infinity Laplacian is zero, equal to $W'(A)$ or $W'(B)$; both viscosity inequalities hold. This argument includes $n = 1$. The identity $(q')^4 = 4W(q)$ extends to

the joins and constant regions, where both sides vanish, proving (6.3). The endpoint limits give the closed range $[A, B]$, so $\mu = 0$. Finally,

$$-T_- = \int_A^{s_*} [4W(s)]^{-1/4}\, ds, \qquad T_+ = \int_{s_*}^B [4W(s)]^{-1/4}\, ds.$$

These nonnegative integrals are understood in the extended sense. Thus the total integral is finite exactly when both endpoints are finite; in that case additivity gives (6.4). □

Corollary 6.2 (Minimum distance between reached phases)**.** *Suppose a solution covered by theorem 1.2 has range in $[A, B]$, attains both endpoints, and $W \geq 0$ on $[A, B]$ with $W(A) = W(B) = 0$. Then any x_A, x_B with $u(x_A) = A$ and $u(x_B) = B$ satisfy*

$$|x_A - x_B| \geq \int_A^B [4W(s)]^{-1/4}\, ds. \tag{6.5}$$

Under the hypotheses of proposition 6.1, this constant is attained when the integral is finite.

Proof. The range inclusion and attainment of both endpoints give $m = A$, $M = B$, and $\mu = 0$. Thus (6.5) follows directly from theorem 4.2. Under the hypotheses of proposition 6.1, finiteness of the total integral makes both T_- and T_+ finite. The points $T_- e$ and $T_+ e$ then attain the values A and B, respectively, and their distance is $T_+ - T_-$, equal to the integral by (6.4). □

6.2. **A smooth example proving the optimal coefficient.**

Proposition 6.3 (Sharpness and failure of the quadratic bound)**.** *For every $n \geq 1$, set*

$$u(x) = \tanh x_1, \qquad W(s) = \frac{1}{4}(1 - s^2)^4. \tag{6.6}$$

Then u is a smooth bounded entire solution of (1.1), with $\mu = 0$, and it attains equality in (1.9) everywhere. Consequently the coefficient $1/4$ in theorem 1.2 cannot be increased. The quadratic inequality $|Du|^2/2 \leq W(u)$ fails.

Proof. Apply proposition 6.1 with $A = -1$, $B = 1$, $s_* = 0$, and $e = e_1$. On $(-1, 1)$, $[4W(s)]^{1/4} = 1 - s^2$, and therefore

$$\Phi(s) = \int_0^s \frac{dt}{1 - t^2} = \operatorname{arctanh} s, \qquad T_- = -\infty, \quad T_+ = +\infty.$$

Thus $q(t) = \tanh t$, and that proposition gives the equation, closed range $[-1, 1]$, normalization $\mu = 0$, and quartic equality. At the origin $|Du| = 1$ and $W(u) = 1/4$. This proves optimality and also gives $|Du(0)|^2/2 = 1/2 > 1/4 = W(u(0))$. □

Remark 6.4 (Why the gradient power is four). If (u, W) solves (1.1), then, for $b > 0$, $u_b(x) = u(bx)$ solves the equation with potential $W_b = b^4 W$ by (2.7). If $Du(x_0) \neq 0$ and $W(u(x_0)) > 0$, the ratio $|Du_b|^p / W_b(u_b)$ satisfies

$$\frac{|Du_b(x_0/b)|^p}{W_b(u_b(x_0/b))} = b^{p-4} \frac{|Du(x_0)|^p}{W(u(x_0))}.$$

For a fixed $p > 4$ this diverges as $b \to \infty$, and for a fixed $p < 4$ it diverges as $b \downarrow 0$. Hence no estimate $|Du|^p \leq CW(u)$ with a finite universal constant C can hold across this family when $p \neq 4$. The example (6.6) at $x_0 = 0$ supplies an original ratio equal to 4, for every fixed p.

6.3. **Equality without one-dimensionality.**

Proposition 6.5 (A smooth radial equality example)**.** *For the standard double-well potential* (4.6)*, the function*

$$u(x) = -\cos|x| \qquad (x \in \mathbb{R}^n) \tag{6.7}$$

is a smooth bounded entire solution of $\Delta_\infty u = u^3 - u = W'(u)$ *and satisfies* $|Du|^4/4 = W(u)$ *everywhere. For* $n \geq 2$*, it is not a function of one linear coordinate.*

Proof. The cosine power series expresses $-\cos|x|$ as a convergent power series in $|x|^2$, proving smoothness at the origin as well as elsewhere. For $r > 0$, lemma 2.3 gives

$$\Delta_\infty u = (\sin r)^2 \cos r = (-\cos r)^3 - (-\cos r) = u^3 - u.$$

The same power series gives $Du(0) = 0$ and $D^2u(0) = \mathrm{Id}$. Since $u(0) = -1$, both sides of the equation are zero there. Also $|Du|^4/4 = \sin^4 r/4 = W(-\cos r)$, including at the origin.

If $u(x) = q(e \cdot x)$ for some unit vector e, it would be constant on the hyperplane $e^\perp$. For $n \geq 2$ there is a unit vector $v \perp e$, but $u(tv) = -\cos|t|$ is not constant, a contradiction. □

Thus no general one-dimensional rigidity statement can be inferred from equality in the quartic estimate, even for smooth bounded entire solutions. Its compatibility with corollary 4.3 follows from the quadratic-well analysis in remark 4.6.

Proposition 6.6 (A compact transition at quadratic wells)**.** *For the potential* (4.6)*, define*

$$u(x) = \begin{cases} -\cos|x|, & |x| \leq \pi, \\ 1, & |x| \geq \pi. \end{cases} \tag{6.8}$$

Then $u \in C^{1,1}(\mathbb{R}^n)$ *is a bounded entire viscosity solution of* $\Delta_\infty u = u^3 - u$*, equals a well outside a ball, and attains the quartic estimate everywhere. The distance* π *from its value* -1 *at the origin to its value* 1 *on the joining sphere is optimal.*

Proof. At $r = \pi$, the inside profile and its first derivative agree with the outside values 1 and 0. Thus composition with $r = |x|$ is C^1 near the joining sphere, and proposition 6.5 gives smoothness at the origin. Explicitly,

$$Du(x) = \frac{\sin|x|}{|x|}x \quad (0 < |x| < \pi), \qquad Du(x) = 0 \quad (x = 0 \text{ or } |x| \geq \pi).$$

Inside the ball, (2.5) gives Hessian eigenvalues $\cos r$ and $\sin r/r$, both of absolute value at most 1; in dimension one only the radial eigenvalue occurs. At the origin $D^2u = \mathrm{Id}$ by proposition 6.5, and outside the ball the Hessian is zero. For $x \neq y$, parametrize the segment by $\gamma(t) = x + t(y - x)$, $0 \leq t \leq 1$. On each open piece separated by its at most two intersections with the joining sphere,

$$\left|\frac{d}{dt}Du(\gamma(t))\right| \leq |y - x|.$$

Integrating on those pieces and using continuity at their endpoints gives $|Du(y) - Du(x)| \leq |y - x|$. The function and its gradient are also bounded, proving the claimed global $C^{1,1}$ regularity.

The equation and equality hold classically away from the joining sphere, by proposition 6.5 and the constant continuation. At any point of that sphere, $Du = 0$ and $W'(1) = W(1) = 0$. The zero-gradient test argument from the proof of proposition 6.1 gives both viscosity inequalities, and the quartic equality there is $0 = 0$. Finally,

$$\int_{-1}^{1} [4W(s)]^{-1/4}\, ds = \int_{-1}^{1} (1 - s^2)^{-1/2}\, ds = \pi.$$

Since $u(0) = -1$ and $u(\pi e) = 1$ for every unit vector e, this distance is attained. Its optimality follows from corollary 6.2. □

7. Scope for boundary data and general forcing

The whole-space hypothesis in theorem 1.2 ensures that, for every fixed $c > 0$, the comparison ball of radius $R = H(M + \delta)$ in lemma 3.1 is available. On its boundary, (3.4) gives $Q_a - u \geq \delta > 0$, excluding boundary contact in lemma 3.2. The spatial limit in (3.8) is taken before $c \downarrow 0$, so no bound on these radii uniform in c is required. On a bounded domain the required ball need not fit; if the comparison meets the domain boundary, the profile must also dominate the boundary data there.

Proposition 7.1 (Arbitrary Dirichlet data exclude a potential-only bound)**.** *Even in a smooth bounded ball, bounded smooth solutions of $\Delta_\infty u = W'(u)$ need not satisfy $|Du|^4/4 \leq W(u)$ for a nonnegative potential.*

Proof. Take $W \equiv 0$ and $u(x) = Kx_1$ on the unit ball, with $K \neq 0$. The function is smooth up to the boundary, satisfies $|u| \leq |K|$, and has the smooth Dirichlet trace $g(x) = Kx_1$ on ∂B_1. Since $D^2u = 0$, it solves the equation classically and hence in the viscosity sense, as explained in the proof of proposition 4.5. Nevertheless,

$$\frac{1}{4}|Du(x)|^4 = \frac{1}{4}K^4 > 0 = W(u(x)) \qquad \text{for every } x \in B_1.$$

□

Accordingly, a Modica estimate on a domain with boundary requires suitable boundary information or a conclusion involving boundary terms. Improving the smoothness of the domain alone cannot remove the counterexample.

For spatial forcing, let $u \in C(\mathbb{R}^n)$ be a bounded viscosity solution of $-\Delta_\infty u = f(x)$ and let $I = [\inf u, \sup u]$. If there are an open interval $J \supset I$ and a potential $W \in C^1(J)$ such that

$$f(x) = -W'(u(x)) \qquad \text{for every } x \in \mathbb{R}^n,$$

then the two equations coincide in the viscosity sense, and theorem 1.2 gives the estimate with $W - \min_I W$. For a general spatial source this factorization is not part of the data; it requires, in particular, that $f(x) = f(y)$ whenever $u(x) = u(y)$. Thus the theorem by itself supplies no gradient–potential conclusion without that additional structure. This observation does not exclude interior gradient estimates involving the source, oscillation, or distance to the boundary.

Finally, the adjective *sharp* in this paper refers to the quartic power and the coefficient $1/4$ established in section 6, and to the profile distance attained in corollary 6.2. The Hölder exponent in theorem 1.3 is an explicit consequence of theorem 1.1; its optimality is not asserted. No unconditional classification of equality cases is stated: proposition 6.5 rules out a general one-dimensional conclusion when $n \geq 2$.

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